

Diploma Thesis

Investigation of Machine Vibration Analysis System Design

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① Introduction

② Background

③ Engineering and Design

④ Implementation

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- Development of a React frontend that requests and visualizes data from the Vercel API.

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Fast-Fourier Transform
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FFT

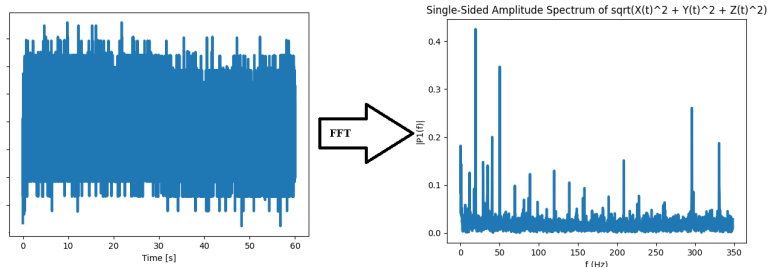


Figure 2: Conversion of the time-domain signal into its frequency-domain power representation using the FFT.

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Machinery Fault Tree Analysis

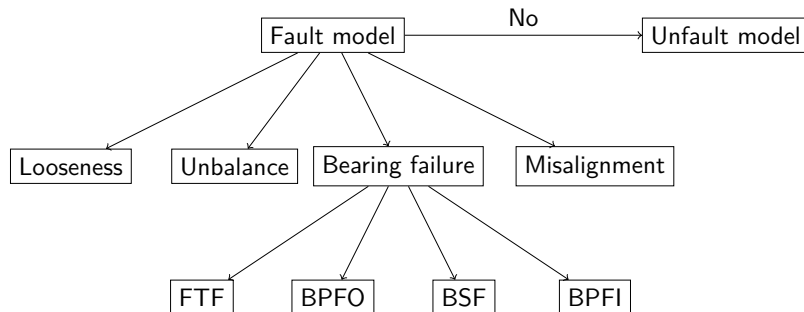


Figure 3: Fault tree analysis diagram showing the relationship between machine conditions and specific fault frequencies.

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System Architecture and Dataflow

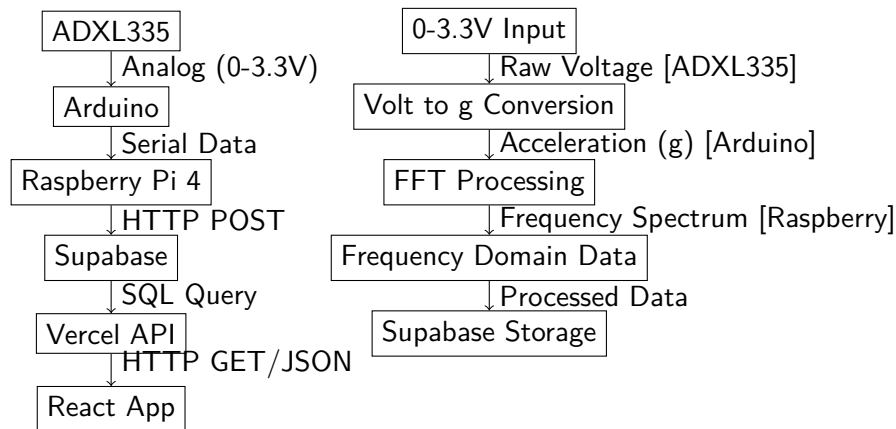


Figure 4: Complete system architecture and data flow and transformation pipeline from physical sensor to web visualization dashboard.

System boundaries

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Raspberry Pi Algorithm

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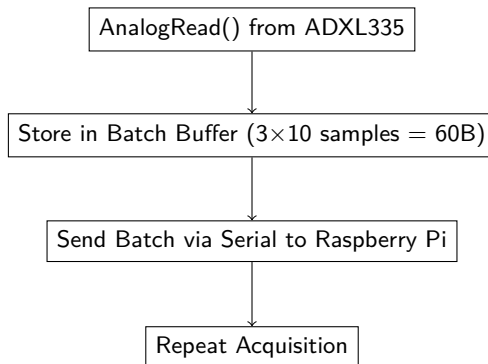


Figure 5: The Arduino UNO Algorithm.

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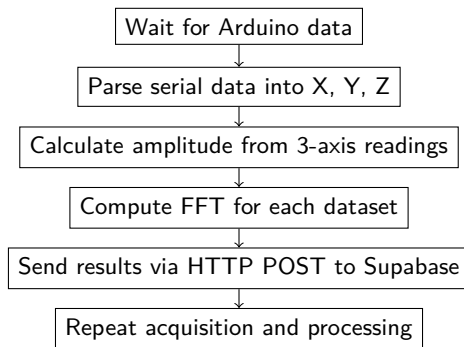


Figure 6: The Raspberry Pi 4 Algorithm: serial data reception, FFT processing, and cloud transmission.

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Vercel and Supabase Algorithmic Flow

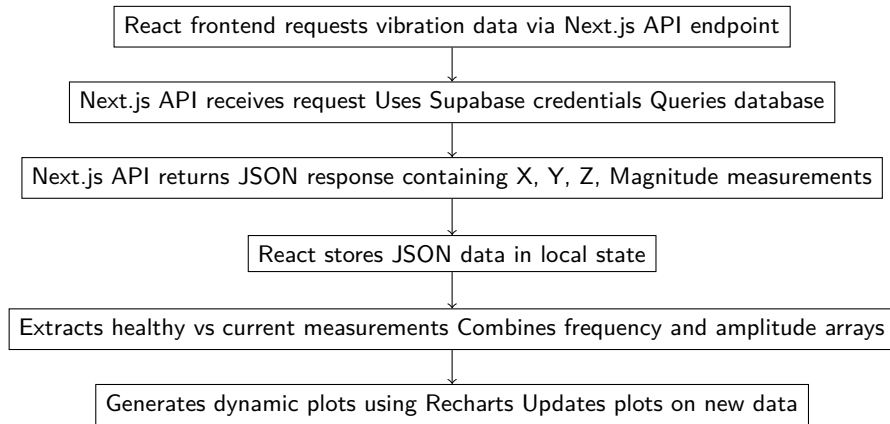


Figure 7: Flow diagram of the Vercel and Supabase algorithm: from frontend request to dynamic visualization.

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System Verification and Validation

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- System performance validated for FFT processing, transmission rates, and visualization updates.

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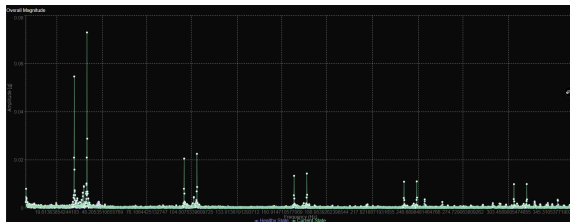


Figure 8: Visualization of the overall vibration magnitude. This chart combines all three coordinate axes into a single representation, showing the total vibration energy with the healthy baseline (blue) compared against the current state (green).

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- Vercel dashboard visualizes frequency spectra interactively for real-time monitoring.